

Class 9: Candy Mini-Project

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Exploratory Analysis of Halloween Candy

```
candy_file <- "https://raw.githubusercontent.com/fivethirtyeight/data/master/candy-power-rankings.csv"
candy=read.csv(candy_file, row.names=1)
head(candy)
```

	chocolate	fruity	caramel	peanutyalmondy	nougat	crispedricewafer
100 Grand	1	0	1	0	0	1
3 Musketeers	1	0	0	0	1	0
One dime	0	0	0	0	0	0
One quarter	0	0	0	0	0	0
Air Heads	0	1	0	0	0	0
Almond Joy	1	0	0	1	0	0

	hard	bar	pluribus	sugarpercent	pricepercent	winpercent
100 Grand	0	1	0	0.732	0.860	66.97173
3 Musketeers	0	1	0	0.604	0.511	67.60294
One dime	0	0	0	0.011	0.116	32.26109
One quarter	0	0	0	0.011	0.511	46.11650
Air Heads	0	0	0	0.906	0.511	52.34146
Almond Joy	0	1	0	0.465	0.767	50.34755

What is in the dataset?

Q1. How many different candy types are in this dataset?

```
nrow(candy)
```

```
[1] 85
```

Q2. How many fruity candy types are in the dataset?

```
sum(candy$fruity)
```

```
[1] 38
```

What is your favorite candy?

```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

```
filter, lag
```

The following objects are masked from 'package:base':

```
intersect, setdiff, setequal, union
```

```
candy |>
  filter(row.names(candy)=="Twix") |>
  select(winpercent)
```

```
      winpercent
Twix    81.64291
```

Q3. What is your favorite candy (other than Twix) in the dataset and what is its winpercent value?

```
candy |>
  filter(row.names(candy)=="100 Grand") |>
  select(winpercent)
```

```
      winpercent
100 Grand  66.97173
```

Q4. What is the winpercent value for "Kit Kat"?

```
candy |>
  filter(row.names(candy)=="Kit Kat") |>
  select(winpercent)
```

```
      winpercent
Kit Kat    76.7686
```

Q5. What is the winpercent value for “Tootsie Roll Snack Bars”?

```
candy |>
  filter(row.names(candy)=="Tootsie Roll Snack Bars") |>
  select(winpercent)
```

```
      winpercent
Tootsie Roll Snack Bars 49.6535
```

skim() This function can give us a quick overview of the dataset.

```
library("skimr")
skim(candy)
```

Table 1: Data summary

Name	candy
Number of rows	85
Number of columns	12
Column type frequency:	
numeric	12
Group variables	None

Variable type: numeric

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
chocolate	0	1	0.44	0.50	0.00	0.00	0.00	1.00	1.00	
fruity	0	1	0.45	0.50	0.00	0.00	0.00	1.00	1.00	

skim_vari- able	n_miss- ing	com- plete_rate	mean	sd	p0	p25	p50	p75	p100	hist
caramel	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
peanutyal- mondy	0	1	0.16	0.37	0.00	0.00	0.00	0.00	1.00	
nougat	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
crispedrice- wafer	0	1	0.08	0.28	0.00	0.00	0.00	0.00	1.00	
hard	0	1	0.18	0.38	0.00	0.00	0.00	0.00	1.00	
bar	0	1	0.25	0.43	0.00	0.00	0.00	0.00	1.00	
pluribus	0	1	0.52	0.50	0.00	0.00	1.00	1.00	1.00	
sugarpercent	0	1	0.48	0.28	0.01	0.22	0.47	0.73	0.99	
pricepercent	0	1	0.47	0.29	0.01	0.26	0.47	0.65	0.98	
winpercent	0	1	50.32	14.71	22.45	39.14	47.83	59.86	84.18	

Q6. Is there any variable/column that looks to be on a different scale to the majority of the other columns in the dataset?

The winpercent variable looks to be the most different from the rest of the dataset.

Q7. What do you think a zero and one represent for the candy\$chocolate column?

A 0 means that there is chocolate in the candy while a 1 means that there is no chocolate in the candy.

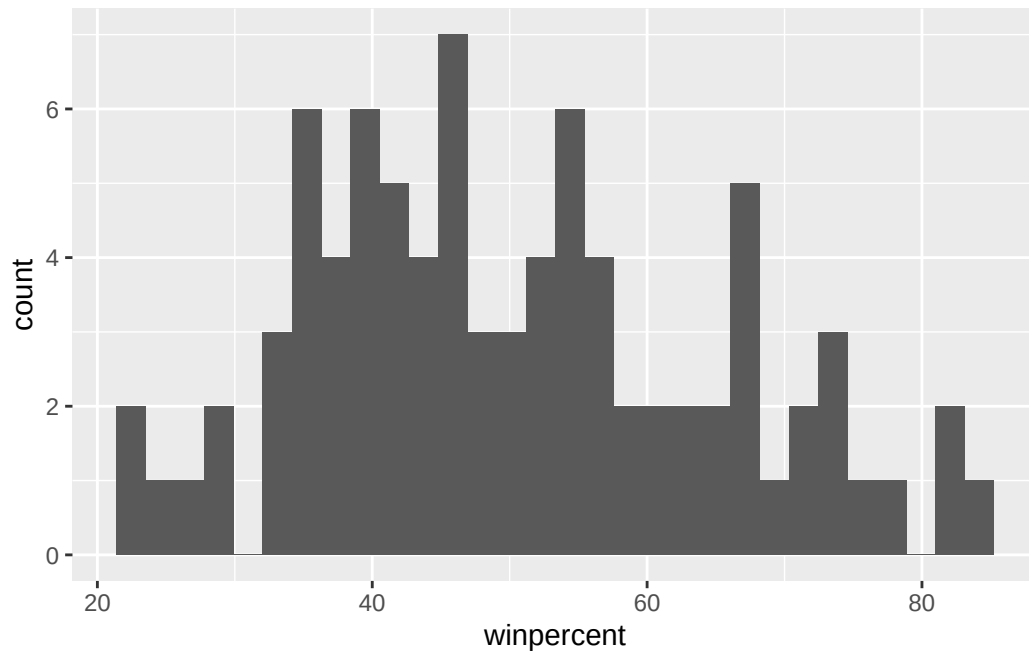
Exploratory analysis

Q8. Plot a histogram of winpercent values using both base R and ggplot2.

```
library(ggplot2)

ggplot(candy, aes(x=winpercent)) +
  geom_histogram()
```

``stat_bin()`` using ``bins = 30``. Pick better value ``binwidth``.



Q9. Is the distribution of winpercent values symmetrical?

No, this distribution of winpercent values is not symmetrical.

Q10. Is the center of the distribution above or below 50%?

```
mean(candy$winpercent)
```

```
[1] 50.31676
```

Q11. On average is chocolate candy higher or lower ranked than fruit candy?

```
mean(candy$winpercent[as.logical(candy$chocolate)])
```

```
[1] 60.92153
```

```
mean(candy$winpercent[as.logical(candy$fruity)])
```

```
[1] 44.11974
```

Chocolate on average is ranked higher than fruity candy.

Q12. Is this difference statistically significant?

```
t.test(candy$winpercent[candy$chocolate==1], candy$winpercent[candy$fruity==1])
```

Welch Two Sample t-test

```
data: candy$winpercent[candy$chocolate == 1] and candy$winpercent[candy$fruity == 1]
t = 6.2582, df = 68.882, p-value = 2.871e-08
alternative hypothesis: true difference in means is not equal to 0
95 percent confidence interval:
 11.44563 22.15795
sample estimates:
mean of x mean of y
 60.92153  44.11974
```

The P-value is less than 0.05 meaning that this data is statistically significant.

Overall Candy Rankings

Q13. What are the five least liked candy types in this set?

```
head(candy[order(candy$winpercent),], n=5)
```

	chocolate	fruity	caramel	peanut	almond	nougat
Nik L Nip	0	1	0		0	0
Boston Baked Beans	0	0	0		1	0
Chiclets	0	1	0		0	0
Super Bubble	0	1	0		0	0
Jawbusters	0	1	0		0	0

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent	price	percent
Nik L Nip				0	0	0	1	0.197		0.976
Boston Baked Beans				0	0	0	1	0.313		0.511
Chiclets				0	0	0	1	0.046		0.325
Super Bubble				0	0	0	0	0.162		0.116
Jawbusters				0	1	0	1	0.093		0.511

	winpercent
Nik L Nip	22.44534
Boston Baked Beans	23.41782
Chiclets	24.52499
Super Bubble	27.30386
Jawbusters	28.12744

Q14. What are the top 5 all time favorite candy types out of this set?

```
head(candy[order(candy$winpercent, decreasing = TRUE),], n=5)
```

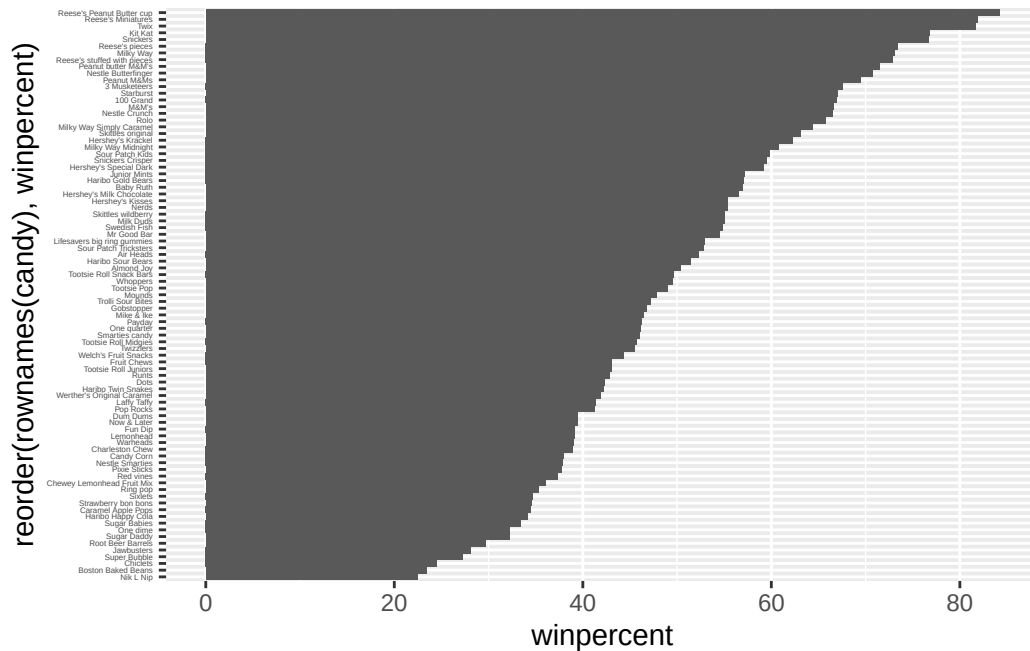
	chocolate	fruity	caramel	peanut	almondy	nougat
Reese's Peanut Butter cup	1	0	0		1	0
Reese's Miniatures	1	0	0		1	0
Twix	1	0	1		0	0
Kit Kat	1	0	0		0	0
Snickers	1	0	1		1	1

	crisped	rice	wafer	hard	bar	pluribus	sugar	percent
Reese's Peanut Butter cup		0	0	0		0		0.720
Reese's Miniatures		0	0	0		0		0.034
Twix		1	0	1		0		0.546
Kit Kat		1	0	1		0		0.313
Snickers		0	0	1		0		0.546

	price	percent	winpercent
Reese's Peanut Butter cup	0.651		84.18029
Reese's Miniatures	0.279		81.86626
Twix	0.906		81.64291
Kit Kat	0.511		76.76860
Snickers	0.651		76.67378

Q15. Make a first barplot of candy ranking based on winpercent values.

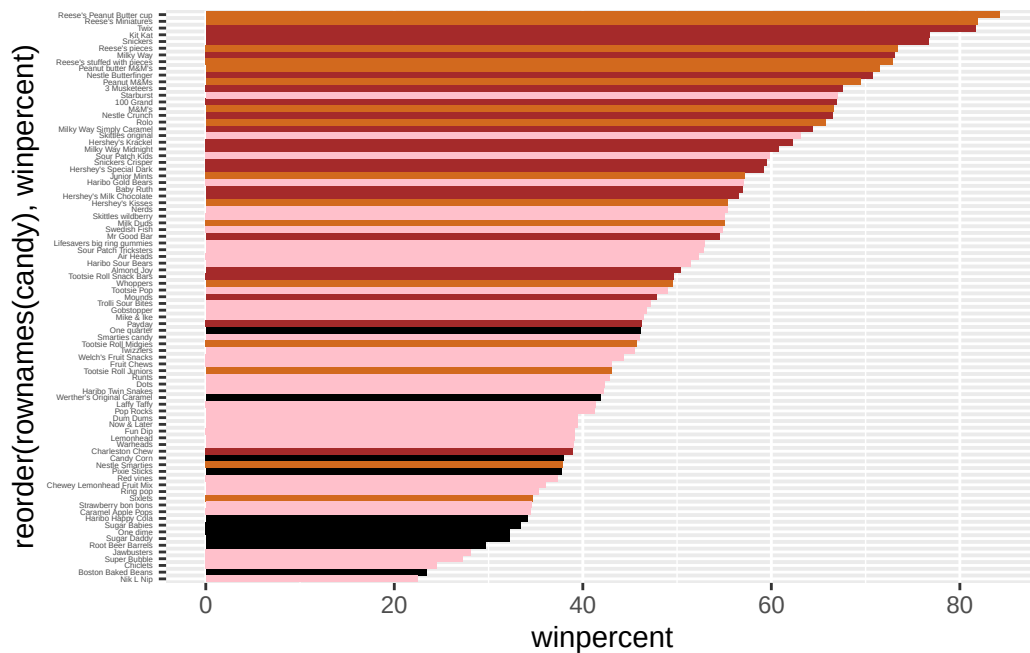
```
ggplot(candy)+
  aes(winpercent, rownames(candy))+
  geom_bar(stat="identity")+
  theme(axis.text.y = element_text(size=3))
```

Time to add some useful color

```
my_cols=rep("black", nrow(candy))
my_cols[as.logical(candy$chocolate)] = "chocolate"
my_cols[as.logical(candy$bar)] = "brown"
my_cols[as.logical(candy$fruity)] = "pink"
```

```
ggplot(candy) +
  aes(winpercent, reorder(rownames(candy),winpercent)) +
  geom_col(fill=my_cols)+
  theme(axis.text.y = element_text(size=3))
```



Q17. What is the worst ranked chocolate candy?

Sixlets is the worst ranked chocolate candy.

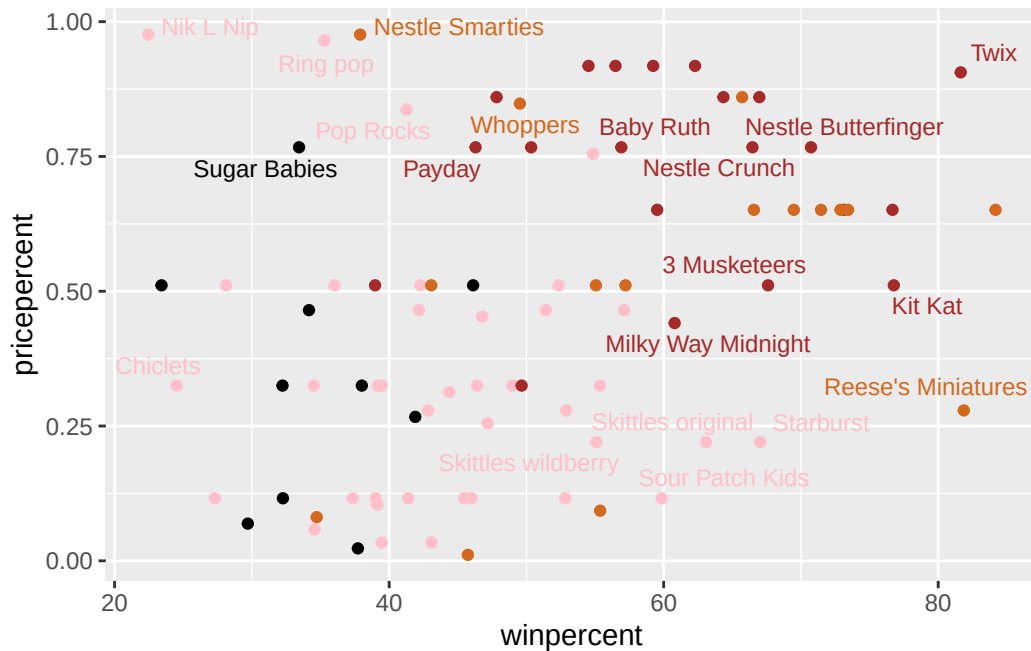
Q18. What is the best ranked fruity candy?

Nik L Nip is the worst ranked fruity candy.

Taking a look at pricepercent

```
library(ggrepel)
ggplot(candy)+
  aes(x=winpercent, y=pricepercent, label=rownames(candy))+
  geom_point(col=my_cols)+
  geom_text_repel(col=my_cols, size=3.3, max.overlaps=5)
```

Warning: ggrepel: 65 unlabeled data points (too many overlaps). Consider increasing max.overlaps



Q19. Which candy type is the highest ranked in terms of winpercent for the least money - i.e. offers the most bang for your buck?

Reese's Miniatures

Q20. What are the top 5 most expensive candy types in the dataset and of these which is the least popular?

```
ord <- order(candy$pricepercent, decreasing = TRUE)
head( candy[ord,c(11,12)], n=5 )
```

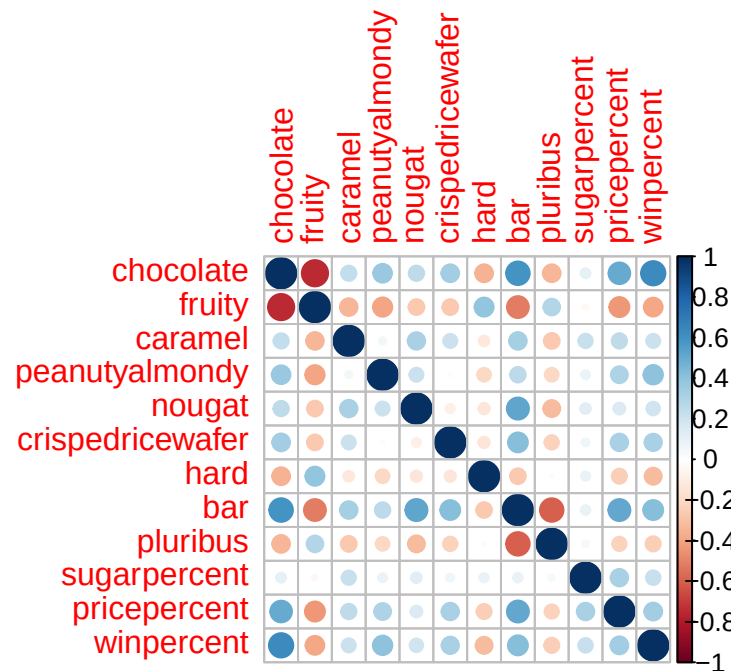
	pricepercent	winpercent
Nik L Nip	0.976	22.44534
Nestle Smarties	0.976	37.88719
Ring pop	0.965	35.29076
Hershey's Krackel	0.918	62.28448
Hershey's Milk Chocolate	0.918	56.49050

Exploring the correlation structure

```
library(corrplot)
```

corrplot 0.95 loaded

```
cij <- cor(candy)  
corrplot(cij)
```



Q22. Examining this plot what two variables are anti-correlated (i.e. have minus values)?

Fruity and chocolate have strong negative correlation.

Q23. Similarly, what two variables are most positively correlated?

Winpercent and chocolate have the most positive correlation.

Principal Component Analysis

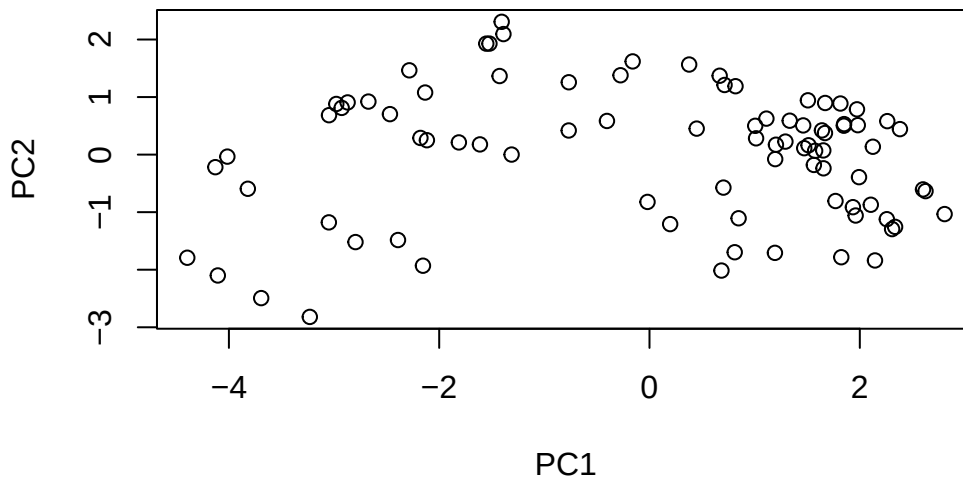
```
pca <- prcomp(candy, scale=TRUE)  
summary(pca)
```

Importance of components:

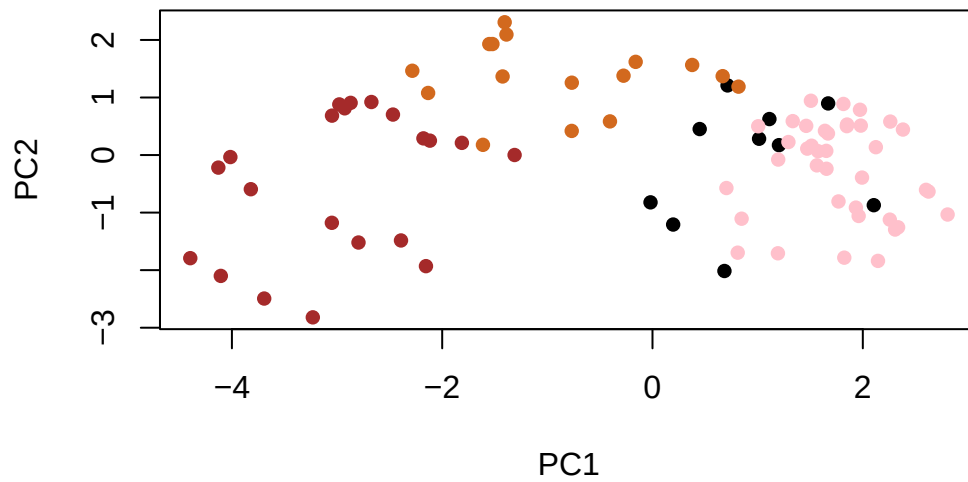
	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	2.0788	1.1378	1.1092	1.07533	0.9518	0.81923	0.81530
Proportion of Variance	0.3601	0.1079	0.1025	0.09636	0.0755	0.05593	0.05539
Cumulative Proportion	0.3601	0.4680	0.5705	0.66688	0.7424	0.79830	0.85369

	PC8	PC9	PC10	PC11	PC12
Standard deviation	0.74530	0.67824	0.62349	0.43974	0.39760
Proportion of Variance	0.04629	0.03833	0.03239	0.01611	0.01317
Cumulative Proportion	0.89998	0.93832	0.97071	0.98683	1.00000

```
plot(pca$x[,1:2])
```



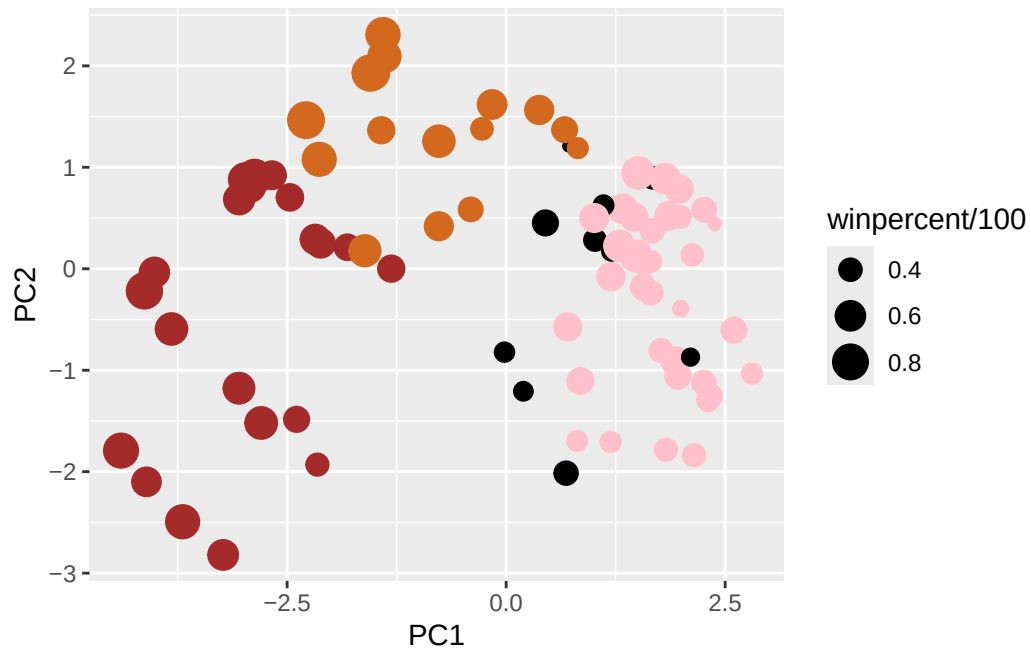
```
plot(pca$x[,1:2], col=my_cols, pch=16)
```



```
my_data <- cbind(candy, pca$x[,1:3])
```

```
p <- ggplot(my_data) +  
  aes(x= PC1, y= PC2,  
      size=winpercent/100,  
      text=rownames(my_data),  
      label=rownames(my_data)) +  
  geom_point(col=my_cols)
```

p



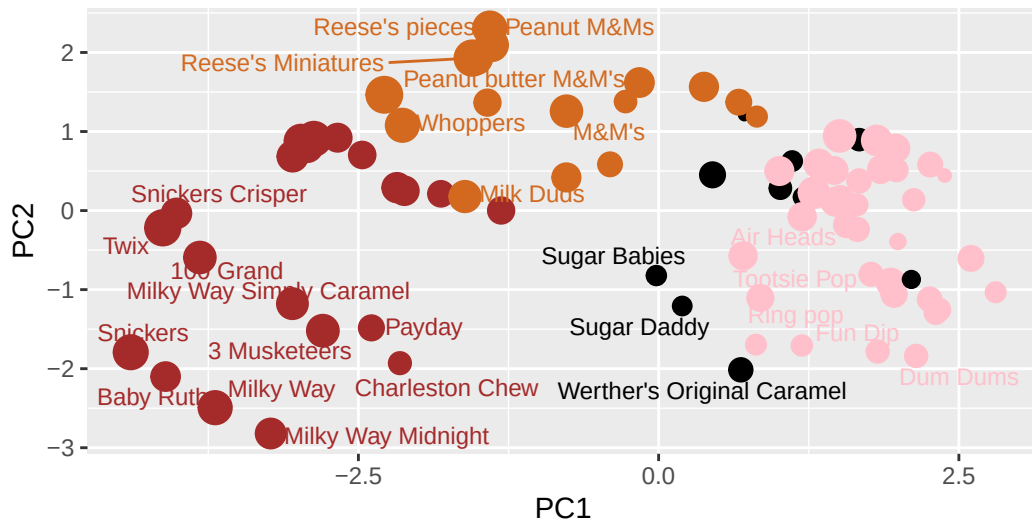
```
library(ggrepel)

p + geom_text_repel(size=3.3, col=my_cols, max.overlaps = 7) +
  theme(legend.position = "none") +
  labs(title="Halloween Candy PCA Space",
        subtitle="Colored by type: chocolate bar (dark brown), chocolate other (light brown),",
        caption="Data from 538")
```

Warning: ggrepel: 59 unlabeled data points (too many overlaps). Consider increasing max.overlaps

Halloween Candy PCA Space

Colored by type: chocolate bar (dark brown), chocolate other (light brown),



Data from 538

```
library(plotly)
```

Attaching package: 'plotly'

The following object is masked from 'package:ggplot2':

last_plot

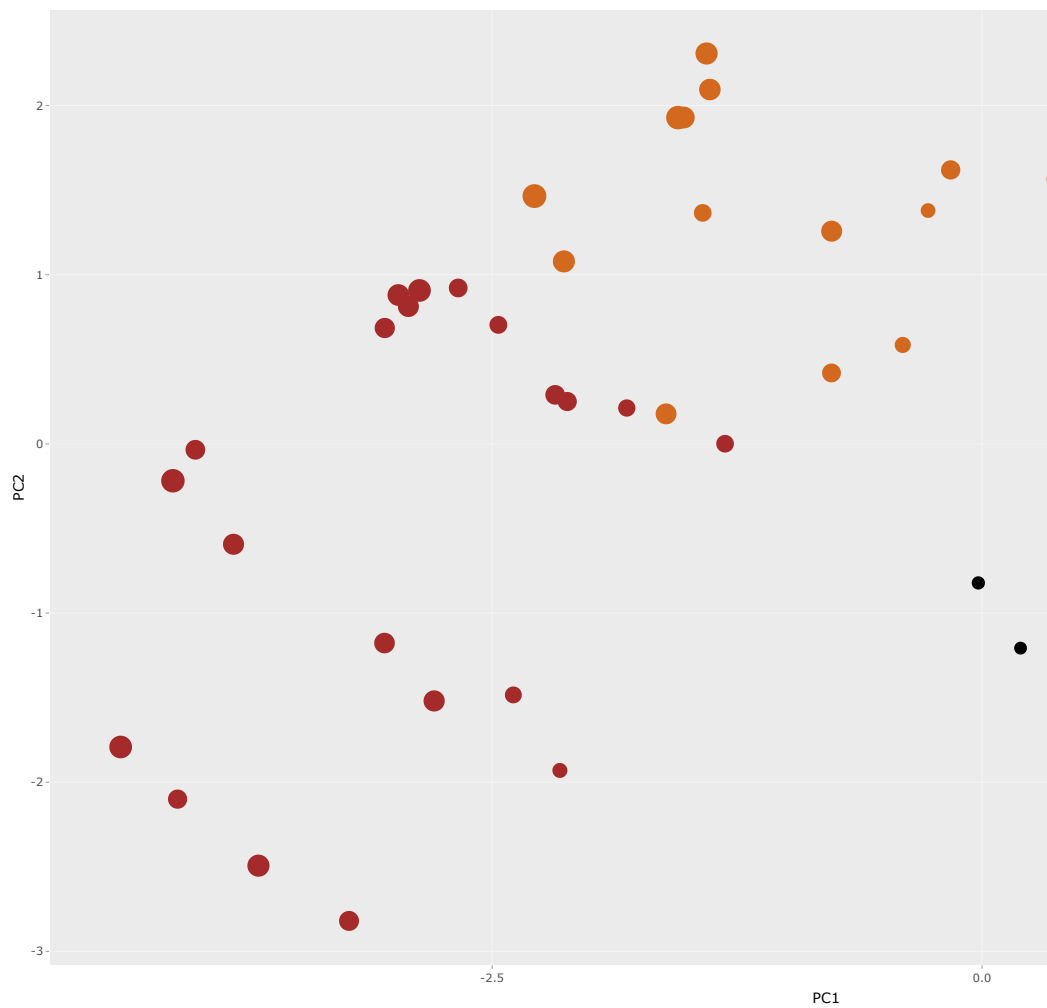
The following object is masked from 'package:stats':

filter

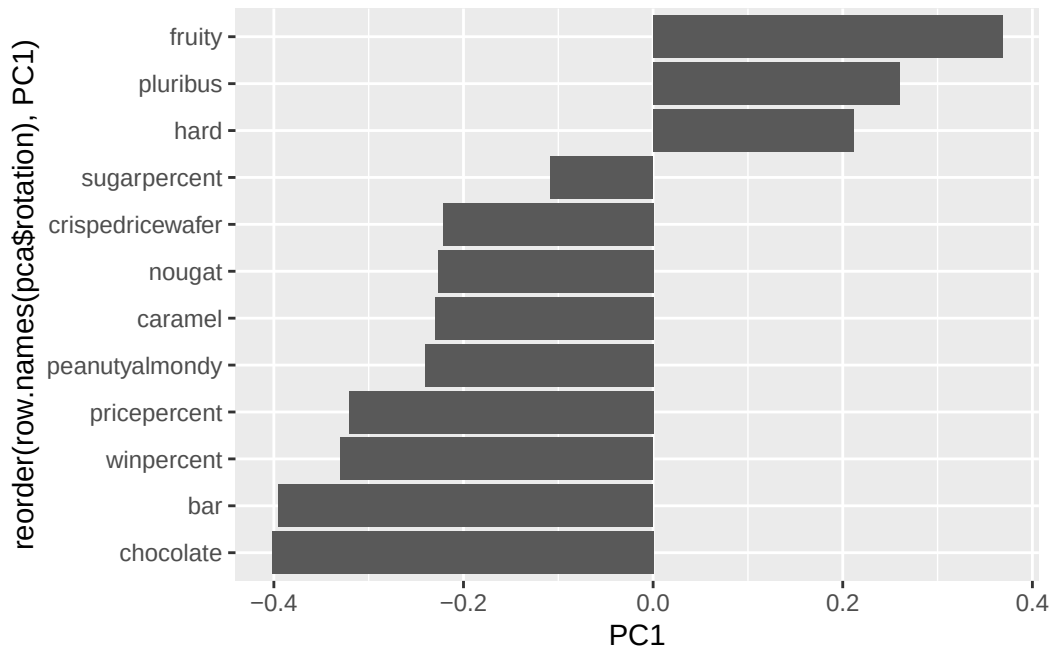
The following object is masked from 'package:graphics':

layout

```
ggplotly(p)
```

```
ggplot(pca$rotation) +
  aes(PC1, reorder(row.names(pca$rotation), PC1)) +
  geom_col()
```



Q24. Complete the code to generate the loadings plot above. What original variables are picked up strongly by PC1 in the positive direction? Do these make sense to you? Where did you see this relationship highlighted previously?

Fruity candy is shown in this graph to be strongly positive which can be also seen in the corplot.